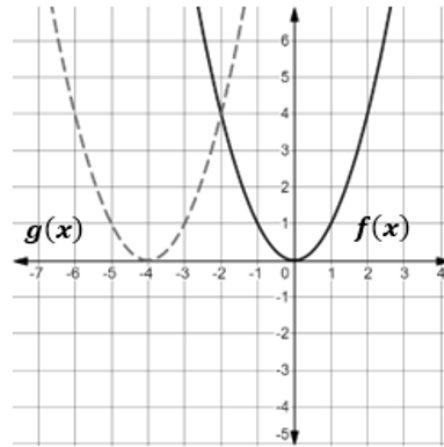


Algebra 1
Unit 13
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

Shifts 4 units to the left

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	-5	1
0	0	-4	0
1	1	-3	1

3. Explain the change in the x -values from $f(x)$ to $g(x)$.

Subtract 4 to each x from $f(x)$ to $g(x)$.

4. Complete the statement:

The graph of $g(x)$ is a ☒ horizontal
☐ vertical shift of $f(x)$.

5. Complete the statement:

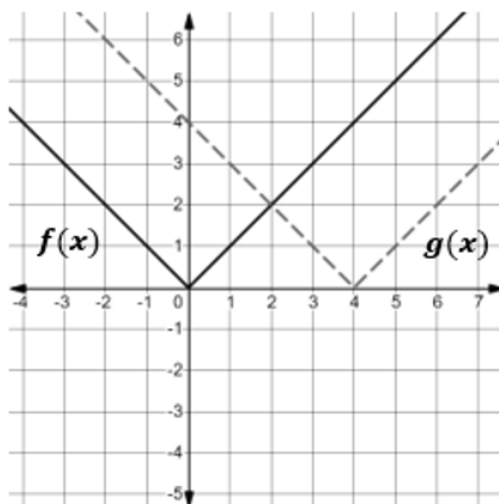
The shift is seen in the table by ☐ adding
☒ subtracting 4 to each ☒ x -value
☐ y -value

for each corresponding ☐ x -value.
☒ y -value.

Complete the table for questions 6-8. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^2$	$f(x) = x^2$	Right 2	$f(x - 2) = (x - 2)^2$
6. $y = x $	$f(x) = x $	Right 4	$f(x - 4) = x - 4 $
7. $y = x^3$	$f(x) = x^3$	Left 4	$f(x + 4) = (x + 4)^3$
8. $y = \sqrt{x}$	$f(x) = \sqrt{x}$	Right 5	$f(x - 5) = \sqrt{x - 5}$

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?

$g(x)$ is 4 units to the right of $f(x)$.

10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.

- ☐ $g(x) = f(x + 4)$
- ☒ $g(x) = f(x - 4)$

The vertex of $f(x)$ moved four units to the right resulting in $g(x)$.